Exploration Insights

In this Summary

- Background info
- Descriptive analysis of numerical variables
- Potential visuals

- Overall, I think because the data is uniform and probably created, I am having a hard time finding relationships between the variables

Overall Background

- 255347 rows and 18 cols
- All the predicting variables are **uniform**
- Columns: `'LoanID', 'Age', 'Income', 'LoanAmount', 'CreditScore','MonthsEmployed', 'NumCreditLines', 'InterestRate', 'LoanTerm', 'DTIRatio', 'Education', 'EmploymentType', 'MaritalStatus', 'HasMortgage', 'HasDependents', 'LoanPurpose', 'HasCoSigner','Default'`
- No missing data
- No duplicates
- DTI (debt to income ration) - (total monthly debt payment/gross monthly income) * 100 → would prefer lower dti

Target Variable

For logistic regression:

- **Default** (0-did not default, 1-did default)
    - Only have 29653 for defaulted vs 225694 for did no default, this will affect how well our model will predict or generalize

For linear regression:

- **InterestRate**
- **LoanAmount**
- **CreditScore**

Predictors/Columns

Numerical Summary and Variables

	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Default
count	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000	255347.000000
mean	43.498306	82499.304597	127578.865512	574.264346	59.541976	2.501036	13.492773	36.025894	0.500212	0.116128
std	14.990258	38963.013729	70840.706142	158.903867	34.643376	1.117018	6.636443	16.969330	0.230917	0.320379
min	18.000000	15000.000000	5000.000000	300.000000	0.000000	1.000000	2.000000	12.000000	0.100000	0.000000
25%	31.000000	48825.500000	66156.000000	437.000000	30.000000	2.000000	7.770000	24.000000	0.300000	0.000000
50%	43.000000	82466.000000	127556.000000	574.000000	60.000000	2.000000	13.460000	36.000000	0.500000	0.000000
75%	56.000000	116219.000000	188985.000000	712.000000	90.000000	3.000000	19.250000	48.000000	0.700000	0.000000
max	69.000000	149999.000000	249999.000000	849.000000	119.000000	4.000000	25.000000	60.000000	0.900000	1.000000

Age

- Has every age from 18-69

Income

- Frequent income: 121985, 69492,117102,149401,12329

Loan Amount

- Frequent amount: 221949, 133724,95419,183627,29290

Credit score

- Freq score: 630, 445, 829, 753, 670

MonthsEmployed

- Freq months: 56, 26, 45, 107, 79

numCreditLines

- Anywhere between 1-4

Interest rate

- Top interest rates: 14.92, 2.25, 4.78, 16.44m 7.30

LoanTerm

- Can have loans of 48, 60, 36, 24, 12

Potential Visuals

